Healthcare Risk Prediction — Machine Learning vs Neural Networks

1. Problem Understanding

Description of the Problem

The objective of this project is to build predictive models to determine whether a patient is at risk of diabetes based on various health-related attributes such as age, Body Mass Index (BMI), exercise habits, smoking status, and blood pressure.This is a binary classification problem, where:

- 0 → No Diabetes
- 1 → Diabetes

Two different approaches are implemented:

- A Machine Learning model (Logistic Regression)
- A Neural Network (Deep Learning model)

The goal is to compare their performance and understand their strengths and limitations.

Importance of Predicting Diabetes Risk

Diabetes is a chronic disease that can lead to severe health complications such as heart disease, kidney failure, and vision loss if not detected early.

- Early prediction helps in:
- Preventive healthcare
- Lifestyle modification
- Reducing medical costs

Thus, predictive models can assist doctors in making data-driven decisions and identifying high-risk patients early.

2. Data Exploration Insights

Key observations from the dataset

- The dataset contains patient health records with multiple features.

- There are no missing values, indicating a clean dataset.
- Most features are numerical, making them suitable for machine learning models.
- The target variable (diabetes) is imbalanced, meaning fewer patients have diabetes compared to those who do not.

Important features influencing prediction

- Age: Older individuals show a higher risk of diabetes.The average age of patients is around 51 years, indicating a middle-aged population where diabetes risk is generally higher.
- BMI (Body Mass Index): Higher BMI is strongly associated with diabetes.The average BMI is around 26, indicating that many patients are in the overweight range, which is a strong risk factor for diabetes.
- Blood Pressure: Elevated blood pressure correlates with higher risk.The average blood pressure is approximately 134, indicating a tendency toward elevated levels, which is associated with diabetes risk.
- Exercise Hours: Lower physical activity is linked to higher diabetes risk.The average weekly exercise is approximately 3 hours, but some patients report no physical activity, which may increase diabetes risk.
- Smoking Status: Smokers may have slightly increased risk.Around 30% of patients are smokers, which could contribute to increased health risks including diabetes.

3. Model Implementation

Explanation of the machine learning model

A Logistic Regression model was implemented to predict the likelihood of diabetes in patients based on various health-related features. Logistic Regression is a supervised machine learning algorithm commonly used for binary classification problems.

Since the target variable (diabetes) has two possible outcomes (0 = No Diabetes, 1 = Diabetes), Logistic Regression is an appropriate and effective choice.

Data Preparation:

Before training the model, the dataset was preprocessed as follows:

- The feature variables (X) were separated from the target variable (y).
- The patient_id column was removed as it does not contribute to prediction.
- The dataset was divided into training and testing sets using an 80:20 split.
- Feature scaling was applied using StandardScaler to normalize the data.

This preprocessing ensures that the model performs efficiently and avoids bias due to different feature scales.

Model Training:

The Logistic Regression model was trained using the training dataset. The model learns the relationship between input features (such as age, BMI, exercise, etc.) and the probability of a patient having diabetes.

Logistic Regression works by applying a mathematical function called the sigmoid function, which converts the output into a probability value between 0 and 1.

Based on this probability:

- Values closer to 1 indicate a higher likelihood of diabetes
- Values closer to 0 indicate a lower likelihood

Prediction:

After training, the model was used to make predictions on the test dataset. The predicted probabilities were converted into binary outcomes (0 or 1) using a threshold (typically 0.5).

Model Evaluation Metrics:

To evaluate the performance of the classification model, several metrics were used, including **Accuracy, Confusion Matrix, Precision, Recall, and F1 Score.** These metrics provide a comprehensive understanding of the model's effectiveness, especially in the presence of class imbalance.

a. **Accuracy**

   Accuracy measures the overall correctness of the model by calculating the proportion of correctly predicted instances out of the total instances.

   Accuracy = (True Positives + True Negatives) / Total Predictions

It indicates how often the model makes correct predictions. However, in datasets with class imbalance, accuracy alone may not be a reliable metric.

- **Confusion Matrix**

  A confusion matrix is a table used to evaluate the performance of a classification model by comparing actual and predicted values.

  Structure:

	Predicted No	Predicted Yes
Actual No	True Negative (TN)	False Positive (FP)
Actual Yes	False Negative (FN)	True Positive (TP)

  It provides a detailed breakdown of correct and incorrect predictions.

  a. True Positives (TP): Correctly predicted diabetic patients

  b. True Negatives (TN): Correctly predicted non-diabetic patients

  c. False Positives (FP): Non-diabetic patients incorrectly predicted as diabetic

  d. False Negatives (FN): Diabetic patients incorrectly predicted as non-diabetic

- **Precision**

  Precision measures the proportion of correctly predicted positive cases out of all predicted positive cases.

  Precision = True Positives / (True Positives + False Positives)

  It indicates how reliable the model's positive predictions are.

- **Recall (Sensitivity)**

  Recall measures the proportion of actual positive cases that are correctly identified by the model.

  Recall = True Positives / (True Positives + False Negatives)

  It reflects the model's ability to identify all diabetic patients.

- **F1 Score**

The F1 Score is the harmonic mean of precision and recall, providing a balance between the two metrics.

F1 Score = 2 × (Precision × Recall) / (Precision + Recall)

It combines both precision and recall into a single metric.

Explanation of the neural network architecture

A basic Artificial Neural Network (ANN) was implemented to predict diabetes risk using patient health data. Neural networks are inspired by the human brain and are capable of learning complex patterns from data through multiple interconnected layers.

The neural network was built using a Sequential model, which is a linear stack of layers. The architecture consists of an input layer, two hidden layers, and an output layer.

- **Input Layer**-The input layer receives all the feature variables such as age, BMI, exercise hours, smoking status, and blood pressure.Number of input neurons is equal to the number of features in the dataset.Each neuron represents one feature

- **Hidden Layers**-Hidden layers learn patterns and relationships between input features.ReLU activation introduces non-linearity, allowing the model to capture complex patterns.These layers transform the input data into more meaningful representations. The model includes two hidden layers:First Hidden Layer 16 neurons,Activation function: ReLU (Rectified Linear Unit).Second Hidden Layer 8 neurons and Activation function: ReLU

- **Output Layer-**Produces output between 0 and 1.Represents probability of a patient having diabetes

Training Process

The model was trained using the training dataset.A portion of the data was used for validation to monitor performance.The model learns by adjusting weights to minimize the loss function over multiple epochs

4. Model Evaluation

The performance of both the Logistic Regression model and the Neural Network model was evaluated using multiple metrics, including **accuracy, confusion matrix, precision, recall, and F1-score.** These metrics provide a comprehensive understanding of model performance, especially considering the

imbalanced nature of the dataset.

Both models achieved an accuracy of approximately 91–92%, which initially suggests good performance. However**, accuracy alone is misleading in this case** because the dataset contains a significantly higher number of non-diabetic patients compared to diabetic patients.

A deeper analysis using the confusion matrix reveals a critical limitation. Both models correctly classified most non-diabetic patients but failed to identify any diabetic patients. This is clearly observed from the **confusion matrix, where the number of True Positives (TP) is zero, and all diabetic cases are classified as False Negatives (FN).**

This issue is further reflected in the classification report, where the precision, recall, and F1-score for the diabetic class are all equal to zero. Among these metrics, recall is the most important in healthcare applications, as it measures the model's ability to correctly identify actual diabetic patients**. A recall value of zero indicates that the model fails completely in detecting patients with diabetes**, making it unsuitable for real-world use despite high accuracy.

The results clearly indicate that both models **are biased toward the majority class (non-diabetic patients)**. This bias is caused by the class imbalance in the dataset, where diabetic cases represent only a small portion of the data.

To better understand model behavior, visualizations such as the confusion matrix and training accuracy graphs (for the neural network) were used. These visualizations help in identifying issues like model bias and poor generalization.

Overall, while both models achieved high accuracy, their inability to detect diabetic patients highlights the importance of using multiple evaluation metrics and addressing class imbalance in predictive modeling.

5. Comparative Analysis

The comparison between the Logistic Regression model and the Neural Network model provides important insights into their performance and applicability.

Both models achieved similar accuracy levels; however, neither model was able to correctly predict diabetic cases due to the imbalanced dataset. This demonstrates that increasing model complexity does not guarantee better performance when underlying data issues are not addressed.

Logistic Regression is a simple, efficient, and interpretable model that assumes a linear relationship between input features and the target variable. It performs

well on structured/tabular data and provides clear insights into feature importance, which is particularly valuable in healthcare applications where interpretability is critical.

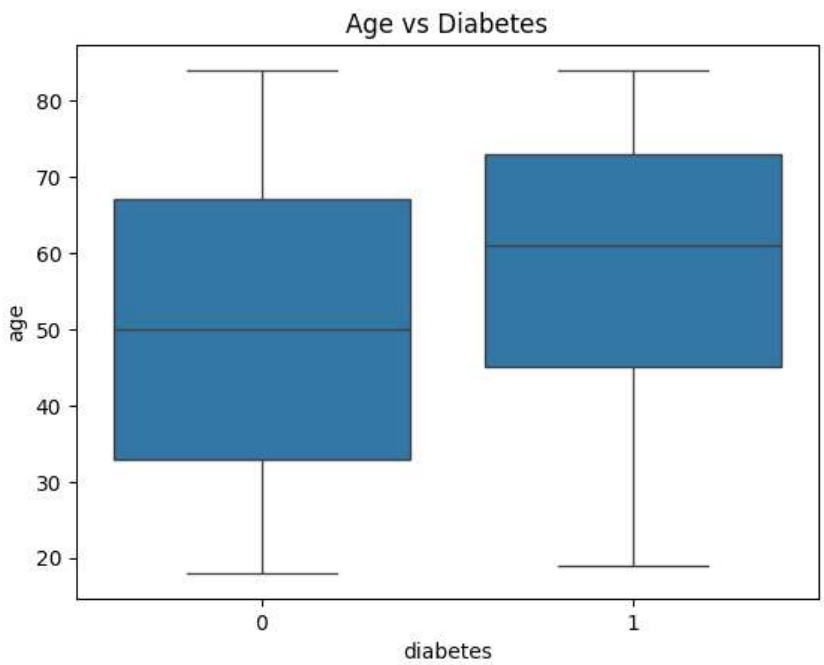
In contrast, the Neural Network model is more flexible and capable of learning complex, non-linear relationships. It uses multiple layers and activation functions to capture deeper patterns in the data. However, neural networks require larger datasets, careful tuning, and are less interpretable, making them harder to apply in sensitive domains like healthcare.

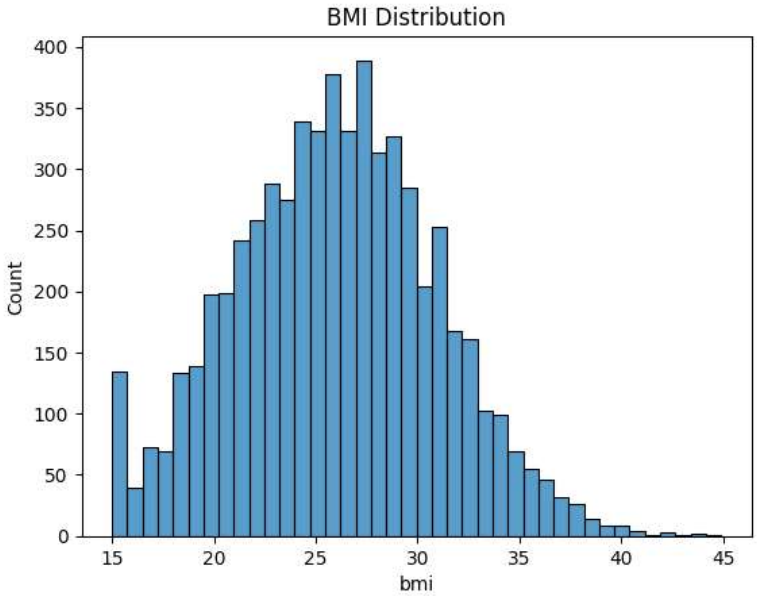
In this study, the neural network did not outperform logistic regression because the main challenge was class imbalance rather than model capability. Both models learned to favor the majority class, leading to poor detection of diabetic patients.

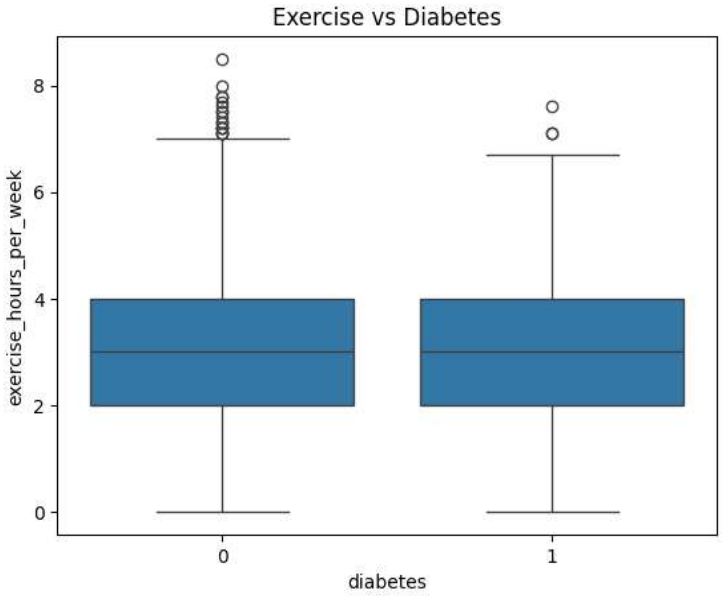
For real-world healthcare applications, Logistic Regression is often preferred due to its simplicity, reliability, and interpretability. However, neural networks can be more effective when working with large datasets and complex patterns, provided that issues like class imbalance are properly handled.

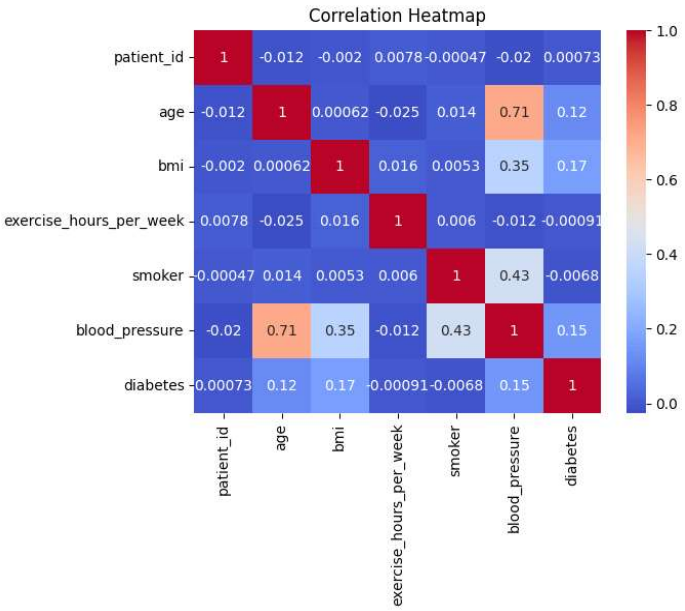
To improve model performance, techniques such as class weighting, oversampling, and synthetic data generation (e.g., SMOTE) can be applied. These methods help the model focus more on the minority class, improving its ability to detect diabetic patients.

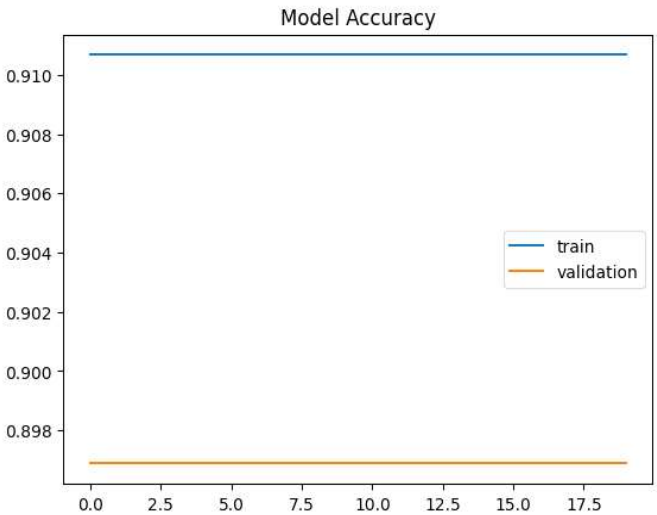
6.Visualisations




BMI Distribution
Count
bmi
0
50
100
150
200
250
300
350
400
15
20
25
30
35
40
45


Exercise vs Diabetes
exercise_hours_per_week
diabetes
0
2
4
6
8
0
1


Correlation Heatmap
patient_id
age
bmi
exercise_hours_per_week
smoker
blood_pressure
diabetes
1 -0.012 -0.002 0.0078 -0.00047 -0.02 0.00073
-0.012 1 0.00062 -0.025 0.014 0.71 0.12
-0.002 0.00062 1 0.016 0.0053 0.35 0.17
0.0078 -0.025 0.016 1 0.006 -0.012 -0.00091
-0.00047 0.014 0.0053 0.006 1 0.43 -0.0068
-0.02 0.71 0.35 -0.012 0.43 1 0.15
0.00073 0.12 0.17 -0.00091 -0.0068 0.15 1
1.0
0.8
0.6
0.4
0.2
0.0


Model Accuracy
0.910
0.908
0.906
0.904
0.902
0.900
0.898
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
train
validation